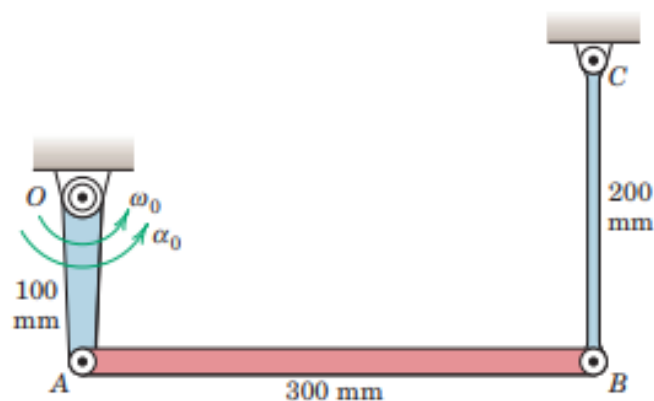
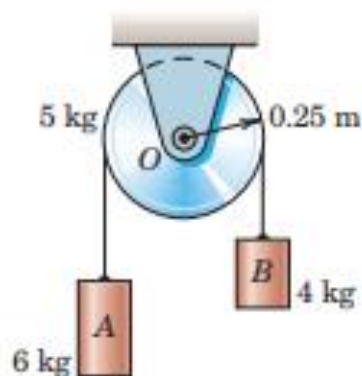


- 6/91** The uniform slender bar AB has a mass of 0.8 kg and is driven by crank OA and constrained by link CB of negligible mass. If OA has an angular acceleration $\alpha_0 = 4\text{ rad/s}^2$ and an angular velocity $\omega_0 = 2\text{ rad/s}$ when both OA and CB are normal to AB , calculate the force in CB for this instant. (*Suggestion: Consider the use of Eq. 6/3 with A as a moment center.*)



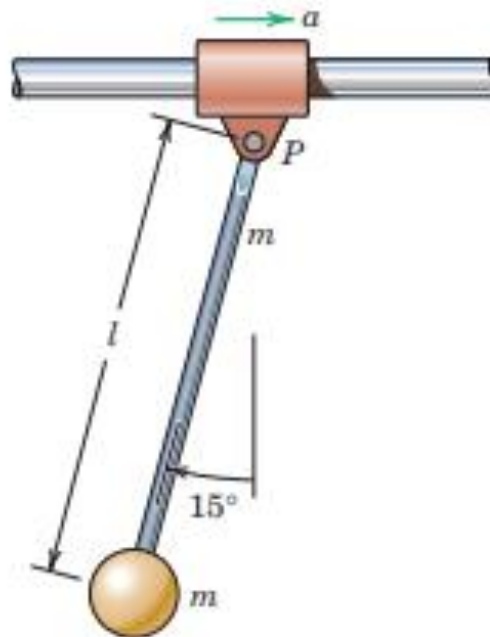
Problem 6/91

- 6/48** Determine the angular acceleration of the uniform disk if (a) the rotational inertia of the disk is ignored and (b) the inertia of the disk is considered. The system is released from rest, the cord does not slip on the disk, and bearing friction at O may be neglected.



Problem 6/48

6/5 What acceleration a of the collar along the horizontal guide will result in a steady-state 15° deflection of the pendulum from the vertical? The slender rod of length l and the particle each have mass m . Friction at the pivot P is negligible.



Problem 6/5

Sugerencia: Intente graficar la posición del péndulo para un rango de tiempo de 5 segundos, use un valor arbitrario para la masa.